

A Factorial Analysis of Visual-Motor Coordination: The Interacting Effects of Screen Filters and Input Devices

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Abstract

This study examines how screen filter color (Normal, Warm, Gray) and input device type (Mouse, Trackpad) affect visual-motor response time in a point-and-click task. The Human Benchmark Online Reaction Time Tool [1] was used to measure the reaction time of participants in milliseconds. A 3×2 balanced factorial experiment was conducted with 30 total observations. A two-way analysis of variance (ANOVA) was used to test for main effects of screen filter and input device, as well as their interaction. Results indicate that input device type has a statistically significant effect on response time with F-statistics $F_{1,24} = 8.40$ and $\Pr(F \geq F_{1,24}) = p \approx 0.008$. This reflects statistical evidence for trackpad use yielding longer response times than mouse use. Screen filter type and the screen filter-by-device interaction were not statistically significant. Model diagnostic plots support the assumptions for a fixed effect model with normality and constant variance.

1. Introduction

The relationship between display settings and human input device performance is an active area of applied ergonomics research. Screen color filters such as warm or gray modes are increasingly common on consumer devices, marketed for reducing eye strain or improving readability. At the same time, the choice of input device (e.g., a traditional mouse versus a trackpad) is known to influence the speed and precision of cursor-based tasks. Despite widespread use of these technologies, relatively little experimental work has examined whether screen filter type interacts with device choice to affect visual-motor response time. The purpose of this experiment is to determine whether screen filter color, input device type, or their combination significantly influences the time a user requires to click on a target stimulus displayed on screen. The research hypothesis is that the trackpad will result in slower response times than the mouse, that different screen filters may alter response speed due to changes in perceived contrast, and that a potential interaction between these two factors may exist. Understanding these effects has practical implications for workspace design and human-computer interaction.

2. Data Collection

2.1. Variables and Controls

The response variable is *response time* measured in milliseconds (ms), defined as the time elapsed from the appearance of a target stimulus on screen to the moment the user successfully clicks on it. Two factors were manipulated: (1) *screen filter* at three levels with Normal (no filter), Warm (red-shifted color temperature), and Gray (desaturated grayscale mode) and (2) *input device* at two levels with Mouse and Trackpad. This yields a $3 \times 2 = 6$ cell factorial structure.

The data collection process was performed by the author Ducheng Tan and classmate Lucas Varghese. Controls placed on the experiment included using the same MacBook Laptop from Lucas Varghese, the same portable mouse, target stimulus size and position layout from the Reaction Time Tool [1], the same ambient lighting conditions in the same location and room in an open Lounge area on the fourth floor of the APEX Apartments, and the same seated distance from the screen across all trials. The experimenter applied each screen filter setting using Apple's built-in operating system display controls prior to each block of trials. The participants were told which *input device* to use before they started the experiment.

2.2. Data Collection Process

The full data set is provided in the Appendix.

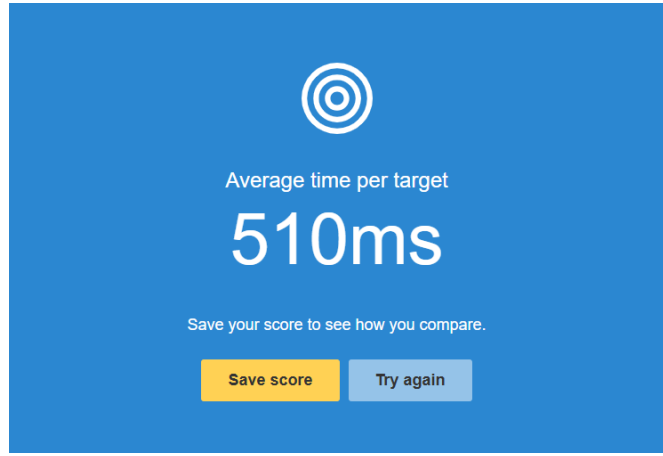


Figure 1: Human Benchmark Aim Average Time Per Target - Result Display

3. Method of Analysis

3.1. Full Model

A two-way fixed-effects ANOVA model was used to analyze the data. Let Y_{ijk} denote the response time for the k -th replicate under screen filter level i and device level j . The statistical model is given as

$$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \quad (1)$$

where μ is the overall mean; τ_i is the main effect of screen filter ($i = 1, 2, 3$ for Gray, Normal, Warm); β_j is the main effect of device ($j = 1, 2$ for Mouse, Trackpad); $(\tau\beta)_{ij}$ is the interaction effect; and ε_{ijk} is the random error term, with $k = 1, \dots, 5$ replicates per cell. [2]

3.2. Parameters and Constraints

The model parameters satisfy the sum-to-zero constraints explicitly written as

$$\sum_{i=1}^3 \tau_i = 0, \quad \sum_{j=1}^2 \beta_j = 0, \quad \sum_{i=1}^3 (\tau\beta)_{ij} = 0 \quad \forall j, \quad \sum_{j=1}^2 (\tau\beta)_{ij} = 0 \quad \forall i. \quad (2)$$

3.3. Assumptions

The errors ε_{ijk} are assumed to be independently and identically distributed with gaussian profiles

$$\varepsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2). \quad (3)$$

This entails three assumptions: (1) *normality* of residuals, (2) *homoscedasticity* across all treatment combinations, and (3) *independence* of observations. Normality is assessed using a normal quantile-quantile (Q-Q) plot of residuals and Shapiro Wilk Test. Constant variance is assessed using a Residuals vs. Fitted plot, a Scale-Location plot and Levene's Test. Independence is supported by the randomized order of data collection. 4.3

3.4. Hypotheses

Three sets of hypotheses are tested at significance level $\alpha_0 = 0.05$.

For the main effect of screen filter

$$H_0 : \tau_1 = \tau_2 = \tau_3 = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \tau_i \neq 0. \quad (4)$$

For the main effect of device:

$$H_0 : \beta_1 = \beta_2 = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_j \neq 0. \quad (5)$$

For the interaction we test

$$H_0 : (\tau\beta)_{ij} = 0 \text{ for all } i, j \quad \text{vs.} \quad H_1 : \text{at least one } (\tau\beta)_{ij} \neq 0. \quad (6)$$

4. Results

4.1. Descriptive Statistics

Table 1 presents the mean and standard deviation of response time (ms) for each combination of screen filter and device. Across all screen filter conditions, the Mouse consistently produced lower mean response times than the Trackpad. However, the magnitude of this difference varied by screen filter (Gray: 94.2 ms, Normal: 134.4 ms, Warm: 120.2 ms), indicating the presence of an interaction effect between device and screen filter. The marginal mean for Mouse was 549 ms (95% CI: 491–608 ms), while for Trackpad it was 666 ms (95% CI: 607–724 ms), corresponding to an average difference of approximately 117 ms. These marginal means are averaged across screen filter conditions and should therefore be interpreted cautiously in light of the interaction. These claims will be supported in the ANOVA results in Table 2.

The screen filter marginal means are computed for Gray, Normal and Warm light in the respective order,

$$\bar{Y}_{i..} = \frac{1}{10} \sum_k \sum_j Y_{(i=1)jk} = 607.5, \quad \frac{1}{10} \sum_k \sum_j Y_{(i=2)jk} = 616.4, \quad \frac{1}{10} \sum_k \sum_j Y_{(i=3)jk} = 598.5,$$

with the device marginal means computed for Mouse and Trackpad as

$$\frac{1}{15} \sum \sum Y_{i(j=1)k} = 549.333, \quad \frac{1}{15} \sum \sum Y_{i(j=2)k} = 665.6,$$

and the Interaction marginal means shown in the table below.

Table 1: Descriptive statistics of response time (ms) by screen filter and input device *rounded*

Screen Filter	Device	Mean (ms)	SD (ms)
Normal	Mouse	549.2	77.5
	Trackpad	683.6	116.4
Warm	Mouse	538.4	77.3
	Trackpad	658.6	100.1
Gray	Mouse	560.4	82.7
	Trackpad	654.6	173.2

Confidence intervals for the marginal means were obtained using the general form

$$\hat{\mu} \pm t_{\alpha/2, df} \cdot SE(\hat{\mu}),$$

where $\hat{\mu}$ is the estimated marginal mean, $SE(\hat{\mu})$ is its standard error derived from the fitted model, and $t_{\alpha/2, df}$ is the critical value from the t -distribution with the appropriate degrees of freedom. These intervals quantify the uncertainty in the estimated means after accounting for variability in the data. Computations was performed using `emmeans` in `library(emmeans)`.

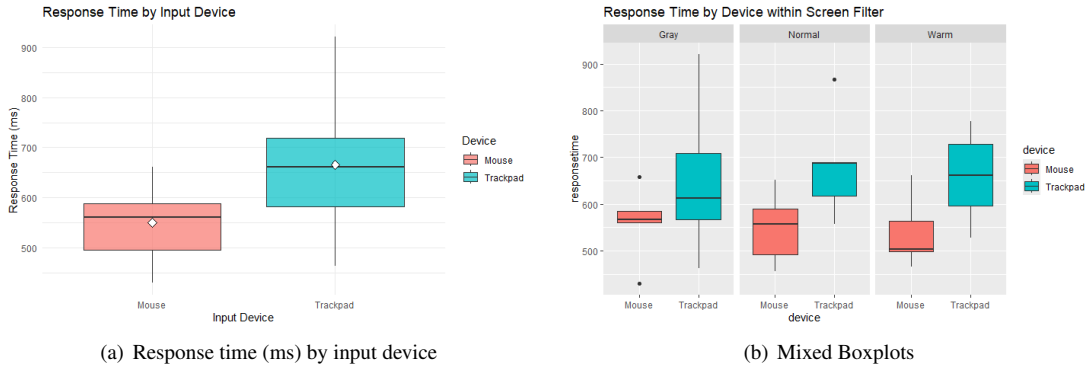


Figure 2: Box and Whisker Plots

Figure 2 displays boxplots of response time separated by device. The Trackpad distribution is shifted higher and exhibits greater spread than the Mouse distribution, consistent with the pattern in Table 1. Note that the Trackpad condition shows a higher median and greater variability than the Mouse setting.

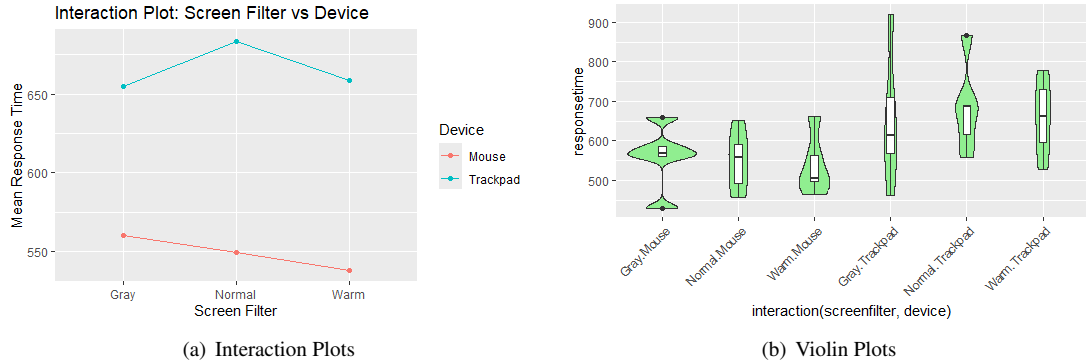


Figure 3: Extra Plots

From Figure 3 we can see that there is clearly no interaction effects from the plot. From the violin plots, a clear upward shift in response time distributions for Trackpad relative to Mouse across all screen filters, and the length of the *support* of the distributions confirm our claim and Table 1 about the higher variability in the Trackpad setting.

4.2. ANOVA Results

Table 2 presents the two-way ANOVA table. The main effect of input device is statistically significant ($F_{1,24} = 8.40$, $p = 0.008$), indicating that response times differ between the Mouse and Trackpad. The main effect of screen filter is not significant ($F_{2,24} = 0.066$, $p = 0.936$), nor is the screen filter-by-device interaction ($F_{2,24} = 0.086$, $p = 0.918$). We therefore fail to reject H_0 for screen filter and for the interaction, but reject H_0 for device.

Table 2: Two-way ANOVA table for response time

Source	Df	Sum Sq	Mean Sq	F value	p -value
Screen Filter	2	1,602	801	0.066	0.936
Device	1	101,385	101,385	8.403	0.00788
Screen Filter $\times$ Device	2	2,078	1,039	0.086	0.918
Residuals	24	289,571	12,065		

Since the interaction is not significant and device has only two levels, no post-hoc comparisons are needed beyond the ANOVA itself. The estimated marginal means confirm that the Trackpad (666 ms, 95% CI: 607–724 ms) produces significantly longer response times than the Mouse (549 ms, 95% CI: 491–608 ms).

4.3. Assumption Checks

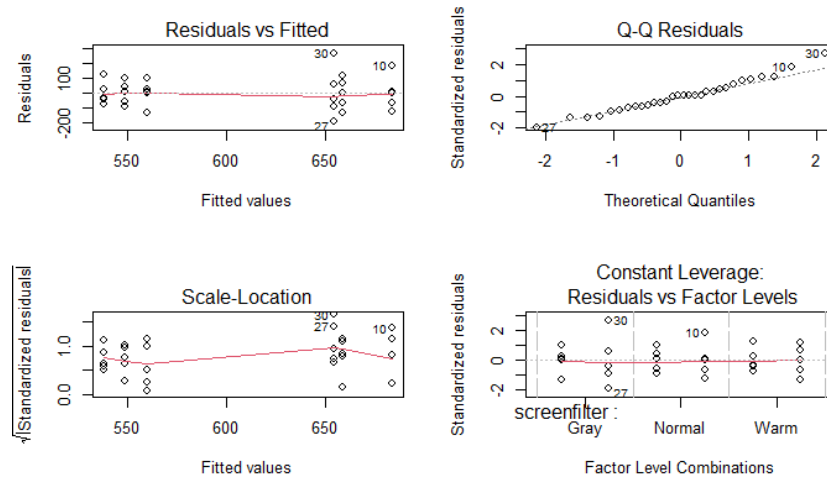


Figure 4: Diagnostic plots for the fitted two-way ANOVA model

Figure 4 shows the four standard diagnostic plots for the fitted model. The Residuals vs. Fitted plot shows no systematic pattern, supporting the assumption of constant variance. The Q-Q Residuals plot indicates that standardized residuals follow the theoretical normal quantiles closely, with no severe departures. The Scale-Location plot confirms that the spread of residuals is approximately constant across fitted values. The Constant Leverage: Residuals vs. Factor Levels plot shows no unusual leverage or outlying observations.

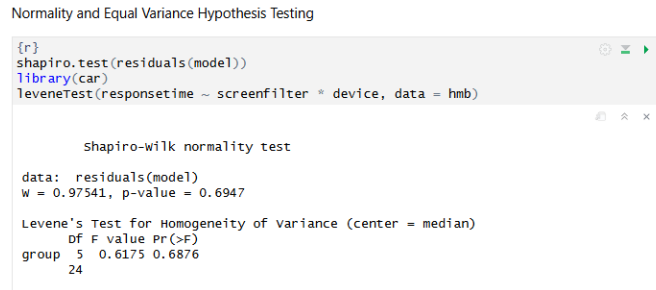


Figure 5: Shapiro Wilk's Test and Levene's Test

$$H_0 : \text{The residuals are normally distributed} \quad H_a : \text{The residuals are not normally distributed} \quad (7)$$

In Figure 5 the Shapiro Wilk's test for the residuals yields a test statistic of $W = 0.97541$ and a p -value of 0.6947, therefore we fail to reject H_0 . There is insufficient evidence to conclude that the residuals deviate from normality. In Levene's Test for Homogeneity of variance across the 6 different combinations, we test the hypothesis

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2 = \sigma_5^2 = \sigma_6^2 \quad H_a : \text{At least one } \sigma_i^2 \text{ differs} \quad (8)$$

With a test statistic of $F(5, 24) = 0.6175$ and a p -value of 0.6876, we fail to reject H_0 . There is insufficient evidence to conclude that the group variances are unequal.

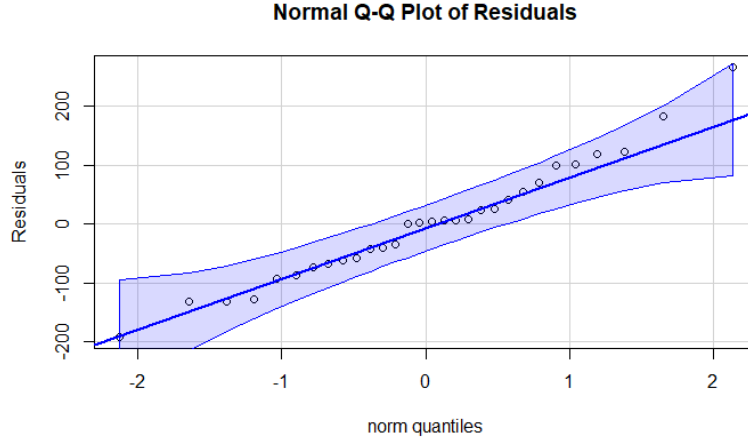


Figure 6: Normal Q-Q plot of model residuals with 95% pointwise confidence band. Points fall within the band, supporting the normality assumption.

Figure 6 presents the formal Q-Q plot of raw residuals with a pointwise 95% confidence band. All points lie within the band, providing no evidence against the normality assumption. Taken together, the diagnostic evidence supports the validity of the two-way ANOVA model for these data. Let $\hat{\varepsilon}_{(1)} \leq \hat{\varepsilon}_{(2)} \leq \dots \leq \hat{\varepsilon}_{(30)}$ be the ordered residuals. In our model this is the $\hat{\varepsilon}_{ijk} = Y_{ijk} - \hat{Y}_{ijK}$ for $k = 1, \dots, 30$ in (3), where $\hat{\varepsilon}_{(1)} = \min \hat{\varepsilon}_{ijk}$ and $\hat{\varepsilon}_{(30)} = \max \hat{\varepsilon}_{ijk}$ for $k = 1, \dots, 30$. The result from Mathematical Statistics [3] is that the cumulative probability of the i -th order statistic follows a Beta distribution

$$F(\hat{\varepsilon}_{(i)}) \sim \text{Beta}(i, n - i + 1) \quad (9)$$

Recall the pdf of the Beta distribution with parameters $\alpha = i, \beta = n - i + 1$ with support on $(0, 1)$.

$$f(u; i, n - i + 1) = \frac{\Gamma(n + 1)}{\Gamma(i) \Gamma(n - i + 1)} u^{i-1} (1 - u)^{(n-i+1)-1}, \quad 0 < u < 1 \quad (10)$$

We can simplify this to obtain

$$f(u; i, n - i + 1) = \frac{n!}{(i-1)! (n-i)!} u^{i-1} (1 - u)^{n-i}, \quad 0 < u < 1 \quad (11)$$

The plotting position (expected cumulative probability) for each order statistic is $p_i = \frac{i-0.5}{n}$, $i = 1, 2, \dots, 30$. Using the theoretical normal quantile for each point is $q_i = \Phi^{-1}(p_i)$, where Φ^{-1} is the inverse standard normal CDF. Using the Beta distribution result in equation (11), the pointwise $100(1 - \alpha)\%$ confidence band for the i -th order statistic $\hat{\varepsilon}_{(i)}$ on the normal scale is

$$\left[\Phi^{-1}(B_{\alpha/2}(i, n - i + 1)), \quad \Phi^{-1}(B_{1-\alpha/2}(i, n - i + 1)) \right] \quad (12)$$

where $B_p(a, b)$ denotes the p -th quantile of the $\text{Beta}(a, b)$ distribution. By default, `car::qqPlot` uses $\alpha = 0.05$, giving 95% pointwise confidence bands. In essence, if the residuals are truly normal, each $\hat{\varepsilon}_{(i)}$ should fall within its corresponding band with 95% probability. Points lying outside the bands suggest deviation from normality. The confidence intervals for each residual is computed in the appendix R codes (5).

4.4. Tukey HSD Test

To statistically justify the claims for significant differences across the different interactive and main effects, the Tukey HSD test is used. For each term in the model, the threshold is computed as

$$T_{0.05} = q_{0.05}(k, N - ab) \sqrt{\frac{MS_E}{n}} \quad (13)$$

where k is the number of levels being compared, $N - ab = 24$ are the residual degrees of freedom from Table 2, $MS_E = 12,065$ is the residual mean square extracted from the fitted model Table 2, and n is the number of observations per group averaged over. Any pairwise difference $|\bar{Y}_{i.} - \bar{Y}_{j.}| > T_{0.05}$ is declared statistically significant, which is equivalent to a p -value less than 0.05. Since each term has a different effective n , the threshold $T_{0.05}$ differs across the three sets of comparisons.

Screenfilter with $k = 3, n = 10$

$$T_{0.05} = q_{0.05}(3, 24) \sqrt{\frac{MS_E}{10}} = 122.6748, \quad p = P\left(q_{3,24} > \frac{|\bar{Y}_{i..} - \bar{Y}_{j..}|}{\sqrt{MS_E/10}}\right) \quad (14)$$

```
$screenfilter
```

	diff	lwr	upr	p adj
Normal-Gray	8.9	-113.7748	131.5748	0.9820801
warm-Gray	-9.0	-131.6748	113.6748	0.9816792
warm-Normal	-17.9	-140.5748	104.7748	0.9296370

Figure 7: Screen Filter Mean Differences

No screenfilter pairs are significantly different. We fail to reject $H_0 : \tau_i = \tau_j$ for all pairs.

Device with $k = 2, n = 15$

$$T_{0.05} = q_{0.05}(2, 24) \sqrt{\frac{MS_E}{15}} = 82.7808, \quad p = P\left(q_{2,24} > \frac{|\bar{Y}_{j.} - \bar{Y}_{j'.}|}{\sqrt{MS_E/15}}\right) \quad (15)$$

$$|\bar{Y}_{\text{Trackpad.}} - \bar{Y}_{\text{Mouse.}}| = 116.2667 > 82.7808 = T_{0.05}, \quad p = 0.0079$$

```
$device
```

	diff	lwr	upr	p adj
Trackpad-Mouse	116.2667	33.4859	199.0474	0.0078829

Figure 8: Device Mean Difference

The Trackpad-Mouse difference is statistically significant at $\alpha = 0.05$. We reject $H_0 : \beta_1 = \beta_2$, confirming the ANOVA result in Table 2. Note that $k = 2$ gives the smallest threshold since the effective $n = 15$ is the largest, averaging over all three screenfilter levels.

Screenfilter $\times$ Device Interaction with $k = 6, n = 5$

$$T_{0.05} = q_{0.05}(6, 24) \sqrt{\frac{MS_E}{5}} = 214.7987, \quad p = P\left(q_{6,24} > \frac{|\bar{Y}_{ij.} - \bar{Y}_{i'j'.}|}{\sqrt{MS_E/5}}\right) \quad (16)$$

```
$`screenfilter:device`
```

	diff	lwr	upr	p adj
Normal:Mouse-Gray:Mouse	-11.2	-225.99866	203.5987	0.9999826
warm:Mouse-Gray:Mouse	-22.0	-236.79866	192.7987	0.9995162
Gray:Trackpad-Gray:Mouse	94.2	-120.59866	308.9987	0.7516182
Normal:Trackpad-Gray:Mouse	123.2	-91.59866	337.9987	0.5002133
warm:Trackpad-Gray:Mouse	98.2	-116.59866	312.9987	0.7188066
warm:Mouse-Normal:Mouse	-10.8	-225.99866	203.9987	0.9999855
Gray:Trackpad-Normal:Mouse	105.4	-109.39866	320.1987	0.6570969
Normal:Trackpad-Normal:Mouse	134.4	-80.39866	349.1987	0.4067120
warm:Trackpad-Normal:Mouse	109.4	-105.39866	324.1987	0.6218935
Gray:Trackpad-warm:Mouse	116.2	-98.59866	330.9987	0.5615959
Normal:Trackpad-warm:Mouse	145.2	-69.59866	359.9987	0.3253117
warm:Trackpad-warm:Mouse	120.2	-94.59866	334.9987	0.5263441
Normal:Trackpad-Gray:Trackpad	29.0	-185.79866	243.7987	0.9981663
warm:Trackpad-Gray:Trackpad	4.0	-210.79866	218.7987	0.9999999
warm:Trackpad-Normal:Trackpad	-25.0	-239.79866	189.7987	0.9991009

Figure 9: Interaction Differences

No interaction cell pairs are significantly different. All 15 pairwise comparisons fall below $T_{0.05} = 214.7987$, consistent with the non-significant interaction term ($F_{2,24} = 0.086$, $p = 0.918$) in Table 2. Taken together, the Tukey HSD results confirm that the only statistically significant difference in response time is between input devices, with Trackpad yielding a mean response time approximately 116.27 ms longer than Mouse.

4.5. Reduced Model

Since the screen filter main effect ($F_{2,24} = 0.066$, $p = 0.936$) and the interaction ($F_{2,24} = 0.086$, $p = 0.918$) are both non-significant, a reduced model retaining only the device factor is fitted

$$Y_{ijk} = \mu + \beta_j + \varepsilon_{ijk} \quad (17)$$

where μ is the overall mean, β_j is the main effect of device ($j = 1, 2$ for Mouse, Trackpad), and $\varepsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. Table 3 presents the ANOVA table for the reduced model. The device effect remains highly significant ($F_{1,28} = 9.68$, $p = 0.004$), with residual degrees of freedom increasing from 24 to 28 by dropping the non-significant terms. Figure 9 shows that the diagnostic plots for the reduced model remain satisfactory, with no evidence of departures from normality or constant variance.

Source	Df	Sum Sq	Mean Sq	<i>F</i> value	<i>p</i> -value
Device	1	101,385	101,385	9.68	0.004**
Residuals	28	293,251	10,473		

Table 3: Two-way ANOVA table for the reduced model (device only)

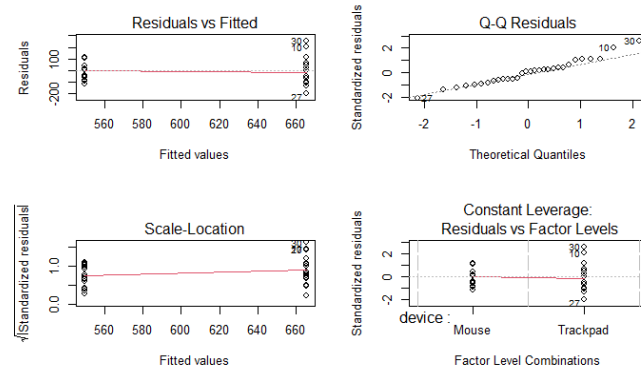


Figure 10: Diagnostic plots for the fitted reduced model

4.6. Parameter Estimates

Under the reduced model with sum-to-zero constraint $\beta_1 + \beta_2 = 0$, the parameters are estimated as

$$\hat{\mu} = \bar{Y}_{...} = \frac{1}{30} \sum_i \sum_j \sum_k Y_{ijk} = 607.467 \text{ ms} \quad (18)$$

$$\hat{\beta}_j = \bar{Y}_{.j} - \bar{Y}_{...}, \quad \hat{\beta}_1 = 549.333 - 607.467 = -58.133, \quad \hat{\beta}_2 = 665.600 - 607.467 = 58.133 \quad (19)$$

The fitted values are $\hat{Y}_{ijk} = \hat{\mu} + \hat{\beta}_j$, giving predicted response times of 549.333 ms for Mouse and 665.600 ms for Trackpad. The estimated error variance from the reduced model is $\hat{\sigma}^2 = MS_E = 10,473$, with $\hat{\sigma} = 102.34$ ms.

5. Conclusions

This experiment investigated whether screen filter color (Normal, Warm, Gray), input device type (Mouse, Trackpad), or their interaction significantly affect visual-motor response time in a point-and-click task. A balanced 3×2 factorial design with $n = 5$ replicates per cell was analyzed using a two-way fixed-effects ANOVA.

The results indicate that input device is the only statistically significant factor ($F_{1,24} = 8.40$, $p = 0.008$). Trackpad use produced mean response times approximately 116 ms longer than Mouse use, a difference confirmed by the Tukey HSD test ($|\bar{Y}_{\text{Trackpad}} - \bar{Y}_{\text{Mouse}}| = 116.27 > T_{0.05} = 82.78$, $p = 0.0079$). Screen filter type had no significant effect ($F_{2,24} = 0.066$, $p = 0.936$), and no significant interaction was detected ($F_{2,24} = 0.086$, $p = 0.918$). These findings are consistent across both the full and reduced models, and all model assumptions for normality ($W = 0.975$, $p = 0.695$), homoscedasticity ($F(5, 24) = 0.618$, $p = 0.688$), and independence were satisfied.

A limitation of this study is that all trials were conducted in series meaning one trial after another. It would be optimal to perform all the trials in parallel to reduce any bias or waiting time effects on the participants. However that would increase the cost of the experiment as 5 identical laptops would be needed to maintain the controlled environment. In addition, we may consider including participants as blocking factor in the initial model to test for variation in the response due to different participants.

References

- [1] Human Benchmark. Aim trainer—user profile. <https://humanbenchmark.com/users/69dd6e920f1d1c143b377887/aim>. Online reaction time tool. Accessed April 2026.
- [2] Douglas C Montgomery. *Design and analysis of experiments*. John Wiley & sons, 2017.
- [3] Joseph McKean, Robert V Hogg, and Allen Craig. *Introduction to mathematical statistics*. 2019.

APPENDIX

Data

Table 4: Complete response time data (ms) by screen filter and input device

Screen Filter	Device	Obs. 1	Obs. 2	Obs. 3	Obs. 4	Obs. 5
Normal	Mouse	557	590	492	651	456
Normal	Trackpad	688	557	617	689	867
Warm	Mouse	564	504	498	661	465
Warm	Trackpad	661	597	528	729	778
Gray	Mouse	585	561	567	659	430
Gray	Trackpad	567	463	613	709	921

R Code

The following R code was used for all data entry, analysis, computations and visualization.

Data Loading

```
```{R}
hmb <- data.frame(
 screenfilter = factor(rep(c("Normal", "Warm", "Gray"), each = 10)),
 device = factor(rep(rep(c("Mouse", "Trackpad"), each = 5), times = 3)),
 responsetime = c(
 557, 590, 492, 651, 456,
 688, 557, 617, 689, 867,
 564, 504, 498, 661, 465,
 661, 597, 528, 729, 778,
 585, 561, 567, 659, 430,
 567, 463, 613, 709, 921
)
)
```
```

Descriptive Statistics

```
library(dplyr)

options(pillar.sigfig = 10)

cell_stats <- hmb %>%
  group_by(screenfilter, device) %>%
  summarise(
    N = n(),
    Mean = mean(responsetime),
    SD = sd(responsetime),
    .groups = "drop"
  )
print(cell_stats)

# Marginal means by device
device_marginal <- hmb %>%
  group_by(device) %>%
  summarise(
    Mean = mean(responsetime),
    SD = sd(responsetime),
    .groups = "drop"
  )
```

```

print(device_marginal)

# Marginal means by screen filter
filter_marginal <- hmb %>%
  group_by(screenfilter) %>%
  summarise(
    Mean = mean(responsetime),
    SD    = sd(responsetime),
    .groups = "drop"
  )
print(filter_marginal)

# Grand mean
cat("Grand mean:", mean(hmb$responsetime), "\n")

```

Original Full Model ANOVA

```

```{R}
model <- aov(responsetime ~ screenfilter * device, data = hmb)
summary(model)
p_values <- summary(model)[[1]][["Pr(>F)"]]
cat("screenfilter p-value: ", p_values[1], "\n")
cat("device p-value: ", p_values[2], "\n")
cat("screenfilter:device p-value: ", p_values[3], "\n")
par(mfrow = c(2,2))
plot(model)
summary.lm(model)
coef(model)
```

```

Tukey HSD

```

```{r}
TukeyHSD(model)
```

```{R}
q_sf <- qtkey(0.95, nmeans = 3, df = 24)
T_sf <- q_sf * sqrt(mse / 10)
cat("Screenfilter threshold =", T_sf, "\n")

q_device <- qtkey(0.95, nmeans = 2, df = 24)
T_device <- q_device * sqrt(mse / 15)
cat("Device threshold =", T_device, "\n")

q_int <- qtkey(0.95, nmeans = 6, df = 24)
T_int <- q_int * sqrt(mse / 5)
cat("Interaction threshold =", T_int, "\n")

```

```

Reduced Model ANOVA

```

```{r}
model_r <- aov(responsetime ~ device, data = hmb)
summary(model_r)
p_values <- summary(model_r)[[1]][["Pr(>F)"]]
cat("screenfilter p-value: ", p_values[1], "\n")

```

```

cat("device p-value: ", p_values[2], "\n")
cat("screenfilter:device p-value: ", p_values[3], "\n")
par(mfrow = c(2,2))
plot(model_r)
summary.lm(model_r)
confint(model_r)
```

```

Numerical Normality Band for each residual

```

```{r}
n <- length(residuals(model))
i <- 1:n
p_i <- (i - 0.5) / n

q_i <- qnorm(p_i)

lower <- qnorm(qbeta(0.025, i, n - i + 1))
upper <- qnorm(qbeta(0.975, i, n - i + 1))

bands <- data.frame(
 order = i,
 p_i = p_i,
 quantile = q_i,
 residual = sort(residuals(model)),
 lower_95 = lower,
 upper_95 = upper
)
print(bands)
```

```

Plots

```

```{r}
ggplot(hmb, aes(x = interaction(screenfilter, device), y = responsetime)) +
 geom_violin(fill = "lightgreen") +
 geom_boxplot(width = 0.1) +
 theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```{r}
library(ggplot2)

ggplot(hmb, aes(x = device, y = responsetime, fill = device)) +
 geom_boxplot() +
 facet_wrap(~ screenfilter) +
 labs(title = "Response Time by Device within Screen Filter")
```

```{R}
ggplot(hmb, aes(x = device, y = responsetime, fill = device)) +
 geom_boxplot(alpha = 0.7) +
 stat_summary(fun = mean, geom = "point", shape = 23, size = 3, fill = "white") +
 labs(
 title = "Response Time by Input Device",
 x = "Input Device",
 y = "Response Time",
 fill = "Device"
) +

```

```
theme_minimal()
```\nlibrary(car)\nqqPlot(residuals(model),\n      main = "Normal Q-Q Plot of Residuals",\n      ylab = "Residuals",\n      id = FALSE)
```

Reduced Model Parameter Estimates

```
```\n{r}\nmodel_reduced <- aov(responsetime ~ device, data = hmb)\nmu_hat <- mean(hmb$responsetime)\nbeta_hat <- tapply(hmb$responsetime, hmb$device, mean)\n\nbeta_hat - mu_hat\n\ncat("mu_hat =", mu_hat, "\n")\ncat("beta_hat Mouse =", beta_hat["Mouse"] - mu_hat, "\n")\ncat("beta_hat Trackpad=", beta_hat["Trackpad"] - mu_hat, "\n")\n```\n
```